

5.2.11. Scatter Plot for Age vs. Fare by S... 05:26

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
assengerid	urvived	class	ame	ex	ge	ibsp	arch	icket	are	abin	mbarked

AgeFareS... Submit

Explorer

```
18 color = {0: 'Red',
19         1: 'Blue'}
20 p = plt.scatter(data['Age'], data['Fare'], c=data['Survived'].map(color))
21
22 plt.title("Age vs. Fare by Survival")
23 plt.xlabel("Age")
24 plt.ylabel("Fare")
25 plt.show()
```

Average time 0.215 s 215.00 ms Maximum time 0.215 s 215.00 ms 1 out of 1 shown test case(s) passed

Age vs. Fare by Survival

Age vs. Fare by Survival

Terminal Test cases

5.1.1. Stacked ... 16:02

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

[Sample Test Cases](#)

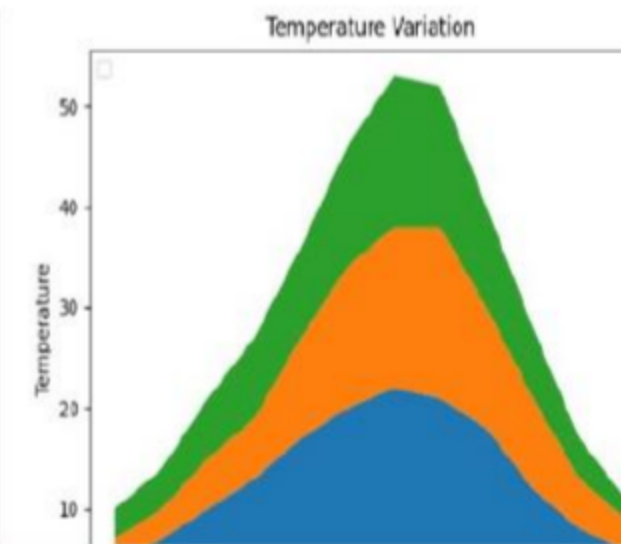
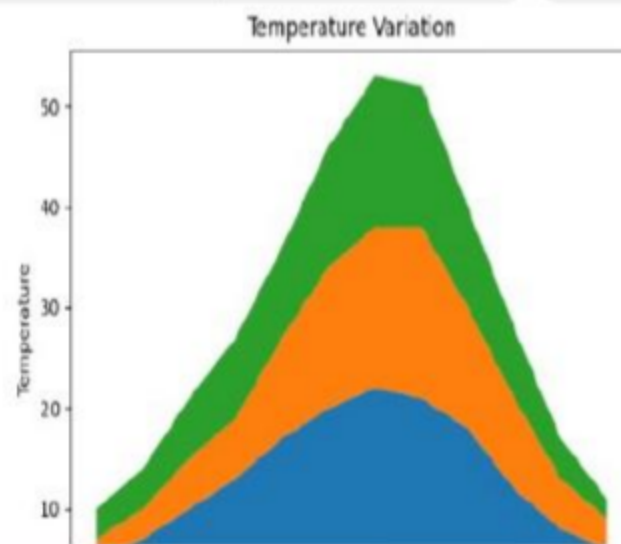
stackedpl...

```
11
12 # Write your code...
13 df = pd.DataFrame(data)
14
15 plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'], df['City_C_Temperature'])
16
17 plt.xlabel('Month')
18 plt.ylabel('Temperature')
19 plt.title('Temperature Variation')
20 plt.legend(loc='upper left')
21 plt.show()
```

Average time
0.193 s
193.00 ms

Maximum time
0.193 s
193.00 ms

✓ 1 out of 1 shown test case(s) passed

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5.2.1. Titanic Dataset

49/37

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

Explorer

titanicDat...

```

43
44
45
46 axes[2,0].scatter(df['Age'],df['Fare'],color=['orange'])
47 axes[2,0].set_title('Fare vs Age')
48 axes[2,0].set_xlabel('Age')
49 axes[2,0].set_ylabel('Fare')
50
51 #plt.subplots_adjust(wspace=7)
52 plt.tight_layout()
53 plt.show()

```

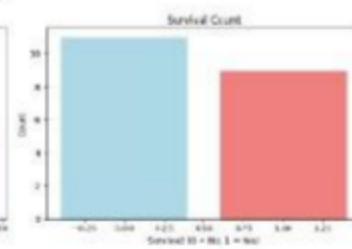
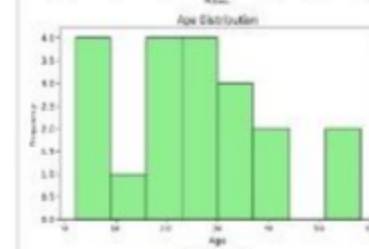
Average time

0.711 s
711.00 ms

Maximum time

0.711 s
711.00 ms

1 out of 1 shown test case(s) passed



Terminal

Test cases

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5.2.2. Histogram of passenger information of Ti... 12:11

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- 1. Use **30 bins** for the histogram.
- 2. Set the **edge color** of the bars to **black (k)**.
- 3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
- 4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	I c k e t	F a r e	C a b i n	E m b a r k e d

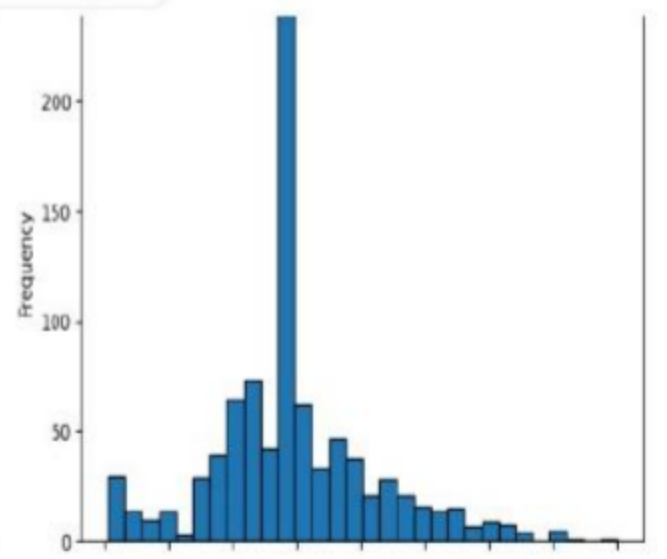
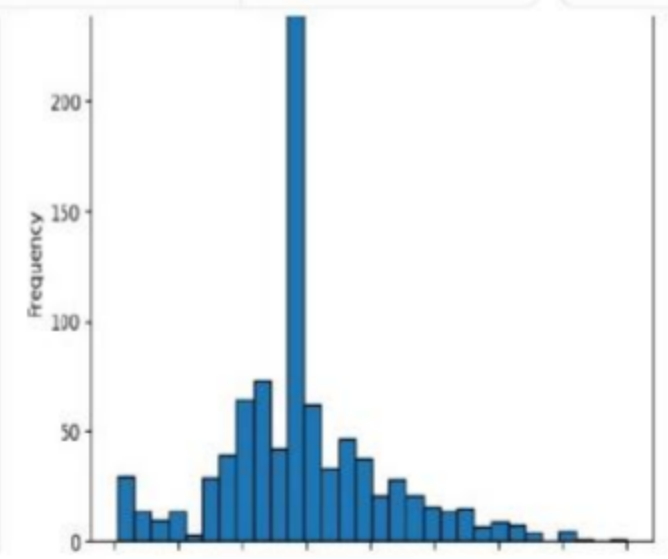
Sample Data:

Histogra...

```
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17
18 plt.hist(data['Age'], bins=30, edgecolor='k')
19
20 plt.xlabel('Age')
21 plt.ylabel('Frequency')
22 plt.title('Age Distribution')
23
24 plt.show()
```

Average time 0.209 s 209.00 ms
Maximum time 0.209 s 209.00 ms

1 out of 1 shown test case(s) passed



Terminal Test cases

5.2.3. Bar plot of survival rate of passengers

04:15



Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset.

The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b s p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

Explorer

BarPlotOf...

```
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17
18 survival_counts = data['Survived'].value_counts()
19 survival_counts.plot(kind = 'bar')
20 plt.title('Survival Count')
21 plt.xlabel('Survived')
22 plt.ylabel('Count')
23 plt.show()
```

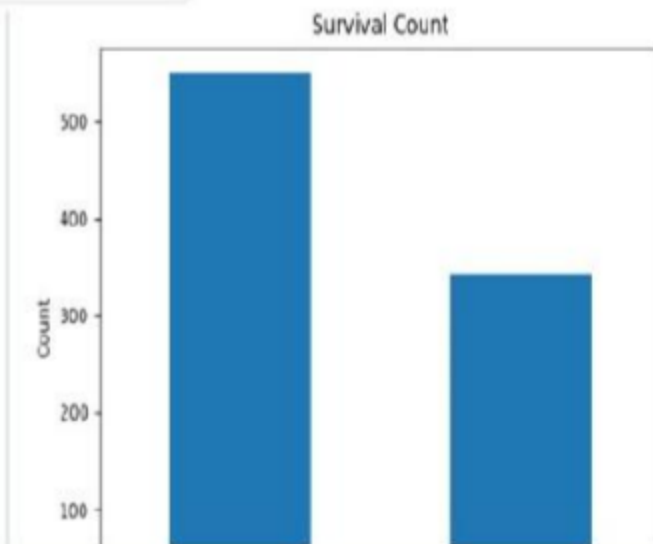
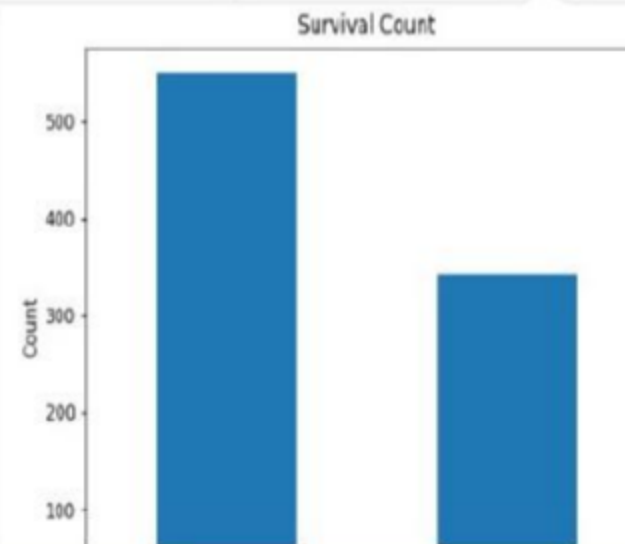
Average time

0.258 s
258.00 ms

Maximum time

0.258 s
258.00 ms

1 out of 1 shown test case(s) passed



Terminal

Test cases

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5.2.4. Bar Plot for Survival by Gender

27:54

- Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:
1. Group the data by the 'Sex' column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
 2. Use a **stacked bar chart** to display the survival counts.
 3. Add the title "Survival by Gender" to the chart.
 4. Label the x-axis as 'Gender' and the y-axis as 'Count'.
 5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	l	a	e	g	i	a	i	a	a	m
s	r	a	m	x	e	b	r	c	r	b	b
e	v	s	e			S	c	k	e	i	n
r	e					p	h	e		n	
i	d										
d											

Explorer

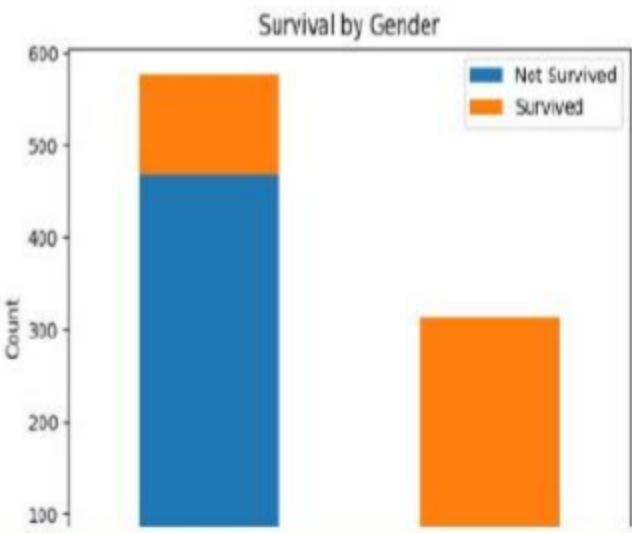
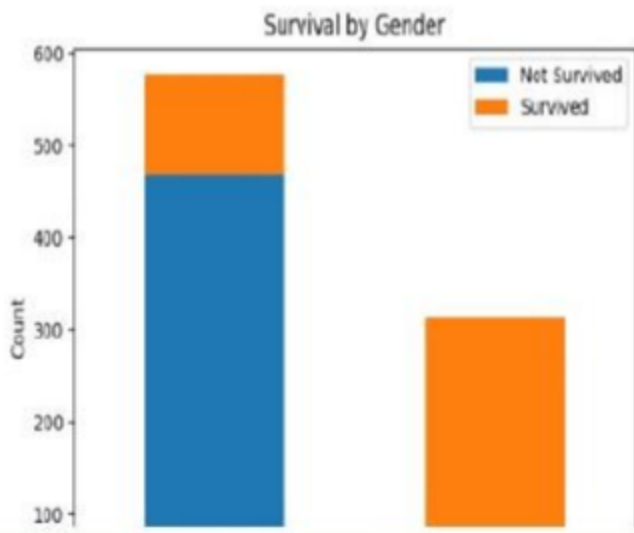
BarPlotOf...

```
16 # Write your code here for Bar Plot for Survival by Gender
17
18 survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
19
20 survival_counts.plot(kind='bar',stacked=True)
21
22 plt.title("Survival by Gender")
23 plt.xlabel("Gender")
24 plt.ylabel("Count")
25 plt.legend(["Not Survived", "Survived"])
26 plt.show()
```

Average time
0.288 s
288.00 ms

Maximum time
0.288 s
288.00 ms

1 out of 1 shown test case(s) passed



Terminal

Test cases

5.2.5. Bar Plot for Survival by Pclass 04:54

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

- 1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().
- 2. Use a stacked bar chart to display the survival counts.
- 3. Add the title "Survival by Pclass" to the chart.
- 4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.
- 5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	c	a	e	g	i	a	i	a	a	m
s	r	l	m	x	e	b	r	c	r	b	b
s	v	a	e			s	c	k	e	i	k
e	i	s				p	h	e		n	e

BarPlotOf...

16 # Write your code here for Bar Plot for Survival by Pclass

17

18 survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack()

19

20 survival_counts.plot(kind='bar',stacked = True)

21

22 plt.title("Survival by Pclass")

23 plt.xlabel("Pclass")

24 plt.ylabel("Count")

25 plt.legend(["Not Survived","Survived"])

26 plt.show()

Average time 0.283 s 283.00 ms

Maximum time 0.283 s 283.00 ms

1 out of 1 shown test case(s) passed

Survival by Pclass

Terminal

Test cases

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5.2.7. Box plot for Age Distribution

07:45

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
ass	urv	class	ame	ex	ge	ib	rch	cket	are	abin	mbarked
enger	ived					sp					
id	d	s	e								

Sample Data:

BoxPlotF...

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Pclass
17 s = data.boxplot(column='Age', by='Pclass')
18 plt.suptitle("")
19 plt.title("Age by Pclass")
20 plt.xlabel("Pclass")
21 plt.ylabel("Age")
22 plt.show()

Average time 0.288 s 288.00 ms
Maximum time 0.288 s 288.00 ms
1 out of 1 shown test case(s) passed

Age by Pclass

Age by Pclass

Terminal Test cases

5.2.8. Box Plot for Age by Survived 03:56

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

- 1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
- 2. Set the title of the plot to **"Age by Survival"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
assengerid	urvived	class	ame	ex	ge	ibsp	arch	icket	are	abin	mbarked

BoxPlotF...

```
16 # Write your code here for Box Plot for Age by Survived
17 data.boxplot(column='Age',by='Survived')
18 plt.suptitle("")
19 plt.title("Age by Survival")
20 plt.xlabel("Survived")
21 plt.ylabel("Age")
22 plt.show()
```

Average time0.286 s286.00 ms

Maximum time0.286 s286.00 ms

1 out of 1 shown test case(s) passed

Age by Survival

Age by Survival

Terminal

Test cases

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- 1. Use the **Pclass** column to group the data for the boxplot.
- 2. Set the title of the plot to **"Fare by Pclass"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P	S	P	N	S	A	S	P	T	F	C	E
a	u	c	a	e	g	i	a	i	a	a	m
s	r	l	m	x	e	b	r	c	r	b	b
s	v	a	e			s	c	k	e	i	r
e	i	s				p	h	e		n	k
r	v										e
i	e										d
d	d										

BoxPlotF...

16

Write your code here for Box Plot for Fare by Pclass

17

s = data.boxplot(column='Fare',by='Pclass')

18

plt.suptitle("")

19

plt.title("Fare by Pclass")

20

plt.xlabel("Pclass")

21

plt.ylabel("Fare")

22

plt.show()

Average time

0.275 s

275.00 ms

Maximum time

0.275 s

275.00 ms

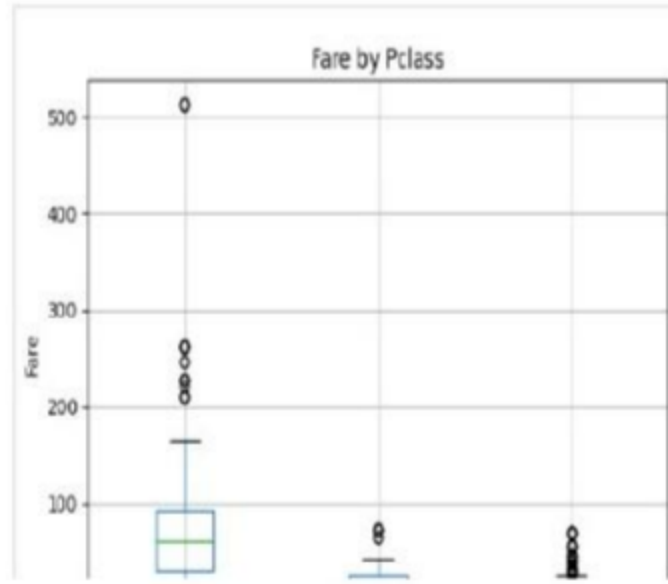
1 out of 1 shown test case(s) passed

Test case 1

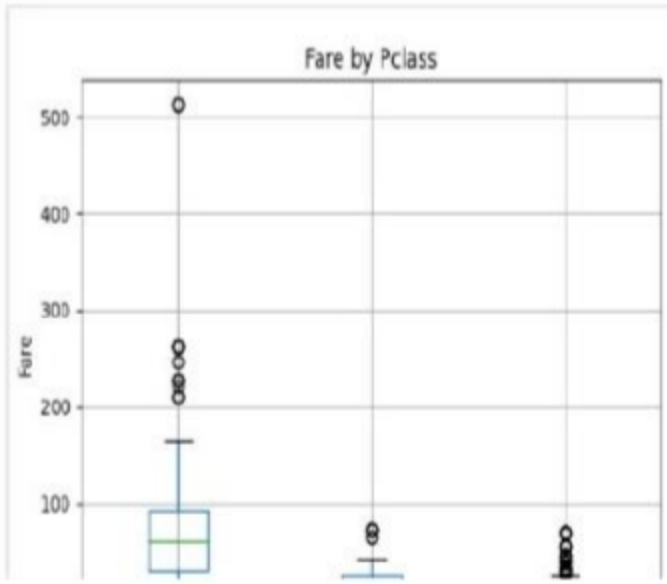
275 ms

Debug

Expected output



Actual output



Terminal

Test cases

5.2.10. Scatter Plot for Age vs. Fare 08:03

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P a s s e n g e r C l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Sample Data:

AgeFareS... Submit

16 # Write your code here for Box Plot for Fare by Pclass
17 p = plt.scatter(data['Age'],data['Fare'])
18 plt.title("Age vs. Fare")
19 plt.xlabel("Age")
20 plt.ylabel("Fare")
21 plt.show()

Average time 0.200 s 200.00 ms Maximum time 0.200 s 200.00 ms 1 out of 1 shown test case(s) passed

Expected output Actual output

Terminal Test cases